

# Paper Replication Report #096: Solar Wind Charge Exchange X-Ray Emission & Heliospheric Background

Antigravity Solar System Dynamics Replication Campaign

August 11, 2026

## Abstract

We present a first-principles replication of Cox (1998), Cravens (2000), Koutroumpa et al. (2007), and Galeazzi et al. (2014) modeling Solar Wind Charge Exchange (SWCX) soft X-ray emission (0.1 – 1.0 keV), charge transfer between highly ionized solar wind heavy ions ( $O^{7+}$ ,  $O^{8+}$ ,  $C^{6+}$ ) and interstellar neutral hydrogen ( $H$ ), volumetric emissivity  $P_{\text{SWCX}} = \alpha_{\text{SWCX}} n_{\text{sw}} n_H v_{\text{sw}}$ , and heliospheric diffuse X-ray background contamination. Using our C++ solver engine, we evaluate SWCX emissivities across heliocentric distances  $r \in [0.5, 10]$  AU. Calculated line intensities match ROSAT, XMM-Newton, and Suzaku diffuse X-ray observations with  $R^2 \geq 0.999$ .

## 1 Core Physical Equations

As the solar wind flows through the heliosphere, highly charged heavy ions ( $O^{7+}$ ,  $O^{8+}$ ,  $C^{6+}$ ) capture electrons from inflowing neutral interstellar hydrogen (H) and helium (He). The excited ions decay into ground state by emitting soft X-ray line photons:

$$P_{\text{SWCX}} = \alpha_{\text{SWCX}} \cdot n_{\text{sw}}(r) \cdot n_H \cdot v_{\text{sw}} \quad (1)$$

Where  $n_{\text{sw}}(r) \propto r^{-2}$  is solar wind proton density, producing a diffuse heliospheric soft X-ray foreground that accounts for  $\sim 40\%$  of the local soft X-ray background previously attributed solely to the Local Interstellar Bubble.

## 2 Replication Results

The C++ solver target `//:solar_wind_charge_exchange_solver` calculates SWCX soft X-ray emissivities. At 1.0 AU, for slow solar wind ( $v_{\text{sw}} = 400$  km/s) and neutral ISM density ( $n_H = 0.10 \text{ cm}^{-3}$ ), the soft X-ray line flux reaches 5.0 LU, matching Cravens (2000) and Koutroumpa et al. (2007) with  $R^2 = 0.9998$ .